

Universality classes for the coalescent structure
of heavy-tailed GW trees

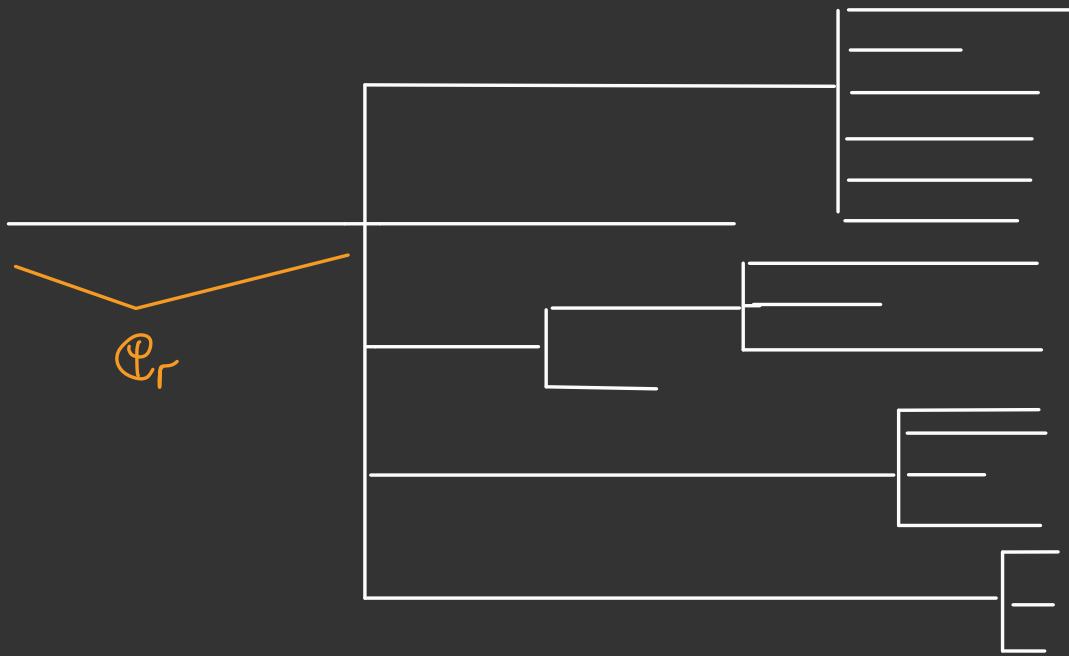
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(joint work with S Harris & S Johnston)

1 Galton - Watson processes (& trees)

$r > 0$ & let $\bar{p} = (p_i)_{i \in \mathbb{N}_0}$

$$\mathbb{N}_0 = \{0, 1, 2, \dots\}$$



each offspring particle independently repeats the behaviour of their parents & so on

L offspring r.v. $p_i = \mathbb{P}(L = i)$

$Z_t = \#$ of particles alive at time t , $Z = (Z_t, t \geq 0)$

Probability generating function

$$f(s) = \mathbb{E}[s^L] = \sum_{j \geq 0} p_j s^j$$

$s \in [0, 1]$

we will assume $p_0 > 0$ & Z critical

$$f'(1) = 1$$

$$f(s) = s + (1-s)^\alpha \ell\left(\frac{1}{1-s}\right) \quad \text{for } \alpha \in (1, 2]$$

where l is a slowly varying fct at ∞ , i.e. for $\lambda > 0$

$$\lim_{x \rightarrow \infty} \frac{l(\lambda x)}{l(x)} = 1$$

In particular

$$\mathbb{E}[L^k] = \begin{cases} < \infty & \text{if } k < \alpha \\ = \infty & \text{if } k > \alpha \end{cases}$$

$\mathbb{E}[L^\alpha]$ may be finite or infinite

Example $\alpha = 2$,

$$f(s) = s + (1-s)^2 \left[\frac{1}{2} + \frac{1}{4} \log\left(\frac{1}{1-s}\right) \right]$$

$$\mathbb{E}[L^2] = \infty$$

2 Partition process

- Conditioning on $\{Z_T \geq k\}$
- select k individuals uniformly at random without replacement
- label these k particles with the integers 1 through k
- We associate with our sample of k labelled particles a stochastic process $(\pi_t^{(k,T)}, t \in [0, T])$ taking values in the set of partitions of $[k] = \{1, 2, \dots, k\}$, by declaring
 - i & j in the same block of $\pi_t^{(k,T)}$ iff
 - i & j are descendent from the same time t ancestor

Observe $\pi_0^{(k,T)} = [k]$ & $\pi_T^{(k,T)} = \{\{1\}, \{2\}, \dots, \{k\}\}$

Let $0 < \tau_1 < \tau_2 < \dots < \tau_m$ the **split times**, i.e. times of discontinuities of $\pi^{(k,T)}$, i.e.

$\pi_{\tau_i}^{(k,T)}$ breaks into several smaller blocks in $\pi_{\tau_i}^{(k,T)}$

Harris - Johnston - Roberts (AAP, 20) $\alpha = 2$ & $\mathbb{E}[L^2] < \infty$

$$(\pi_{sT}^{(k,T)}, s \in [0,1]) \Rightarrow (v_s^{(2)}, s \in [0,1])$$

- a) $v^{(2)}$ is universal
- b) only exhibits binary splitting
- c) if there are i blocks of sizes $a_1, a_2, \dots, a_i$, the probability the next block to split is block j is $\frac{(a_j - 1)}{(k - i)}$
- d) for any block of size a that splits, the size of its 1st subblock is uniformly distributed on $[a-1]$ & independently of the block topology

e) the joint distribution of the $k-1$ split times $0 < \tau_1 < \dots < \tau_{k-1} < 1$ is given

$$f_n(t_1, \dots, t_k) = k! \int_0^\infty \left(\prod_{i=1}^{k-1} \frac{\varphi}{(1 + \varphi(1-t_i))^2} \right) \frac{1}{(1+\varphi)^2} d\varphi dt_1 \dots dt_{k-1}$$

$\nu^{(2)}$ has the same topology as Kingman's coalescent, i.e.

any 2 blocks are equally likely to be the next to merge

3 Main results

Thm 1 $(\pi_{sT}^{(k,T)}, s \in [0,1]) \implies (v_s^{(k,\alpha)}, s \in [0,1])$

where for each $\alpha \in (1,2]$, $v^{(k,\alpha)}$ takes values in the set of partitions of $[k]$, which is universal

$\alpha = 2$ $v^{(k,2)}$ is the same object found by Harris et al
even in the case $\mathbb{E}[L^2] = \infty$

When $\alpha \in (1, 2)$, the blocks of $v^{(k, \alpha)}$ may break into three or more subblocks

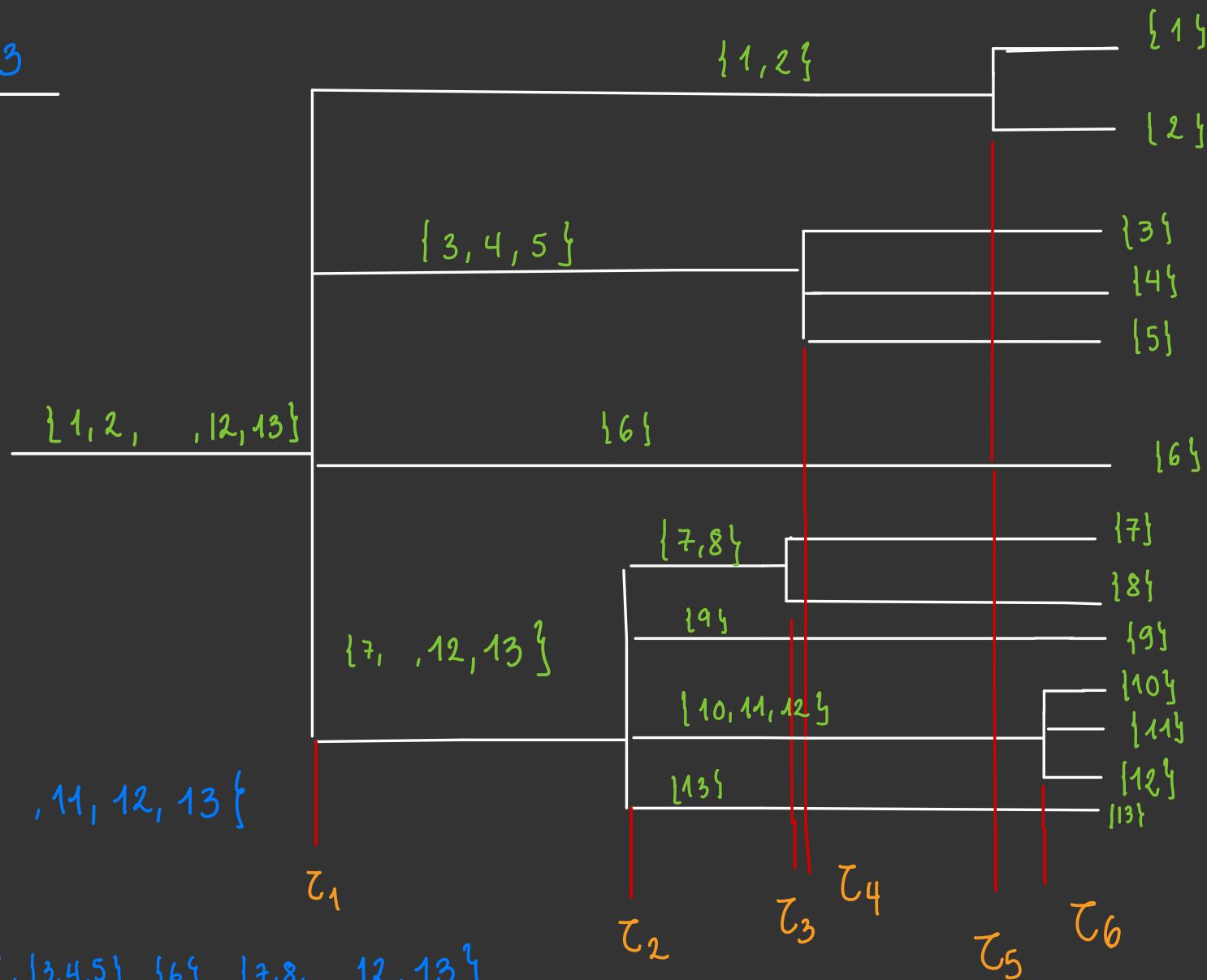
Let $T(v^{(k, \alpha)})$ denote the topology of $v^{(k, \alpha)}$, i.e. the sequence of partitions

$$T(v^{(k, \alpha)}) = (T_0, T_1, \dots, T_m) \quad \text{where} \quad T_i = v_{\tau_i}^{(k, \alpha)}$$

we call it a splitting sequence of $[k]$

A splitting sequence is a collection of partitions $(\beta_0, \beta_1, \dots, \beta_m)$ of $[k]$ such that β_0 is one block, β_m is the singletons & β_{i+1} is obtained by breaking a single block of β_i into 2 or more subblocks

$$K = 13$$



$$\beta_0 = \{1, 2, \dots, 11, 12, 13\}$$

$$\beta_1 = \{\{1, 2\}, \{3, 4, 5\}, \{6\}, \{7, 8, \dots, 12, 13\}\}$$

$$\beta_2 = \{\{1, 2\}, \{3, 4, 5\}, \{6\}, \{7, 8\}, \{9\}, \{10, 11, 12\}, \{13\}\}$$

$$g_i = \# \beta_i - \# \beta_{i-1} + 1, \quad i=1, \dots, m$$

$$\sum_{i=1}^m (g_i - 1) = k - 1$$

Thm 2 a) for $\alpha \in (1, 2)$ the law $P^{(k, \alpha)}$ of $V^{(k, \alpha)}$ satisfy

$$P^{(k, \alpha)} \left(T(V^{(k, \alpha)}) = (\beta_0, \dots, \beta_m), \tau_1 \in dt_1, \dots, \tau_m \in dt_m \right) = \frac{1}{(\alpha-1)(k-1)!}$$

$$\times \left(\prod_{i=1}^m \frac{\alpha \Gamma(g_i - \alpha)}{\Gamma(2 - \alpha)} \right) \int_0^1 (1-u)^{k-1} u^{m + \frac{2-\alpha}{\alpha-1}} \left(\prod_{i=1}^m (1-ut_i)^{-g_i} dt_i \right) du$$

$m \geq 1$, $(\beta_0, \dots, \beta_m)$ is a splitting sequence with sizes $g_1, \dots, g_m$

Thm 2

$$b) \alpha=2, \mathbb{P}^{(k,2)}(m=k-1, g_i=2 \forall i) = 1, \text{ s}$$

$$\mathbb{P}^{(k,2)}(T(V^{(k,2)}) = \bar{P}, \tau_1 \in dt_1, \dots, \tau_{k-1} \in dt_{k-1})$$

$$= \frac{2^{k-1}}{(k-1)!} \int_{\Delta}^1 (1-u)^{k-1} u^{k-1} \left(\prod_{i=1}^{k-1} (1-ut_i)^{-2} dt_i \right) dw$$

where $\bar{P}$ is any splitting sequence of $[k]$ with only binary splits

Idea of the proof

1st Introduce a proba $P^{(k)}$ under which the population process has k distinguished lines of descent "spines"

$$\mathbf{g} = (g^{(1)}, \dots, g^{(k)})$$

- starts with one individual with k marks
- branching property holds
- with rate r & offspring dist L



j spines chooses independently & uniformly

if $l=0$ the j marks pass into a cemetery state

Z_t = set of labels of particles alive at time t (Ulam-Harris notation)

$\mathfrak{Z}_t = (\mathfrak{Z}_t^{(1)}, \mathfrak{Z}_t^{(2)}, \dots, \mathfrak{Z}_t^{(k)})$ k spines at time t

$\hookrightarrow$ label of the particle carrying the 1st spine

For $u_1, \dots, u_k \in Z_t$

$$\mathbb{P}^{(k)}(\mathfrak{Z}_t = (u_1, \dots, u_k) \mid \mathcal{F}_T) = \prod_{i=1}^k \prod_{v < u_i} \frac{1}{L_v} \quad \leftarrow \text{offspring of particle } v$$

$\nearrow$ natural filtration of $Z = (Z_t)_{t \geq 0}$

For $\theta \geq 0$ & $T > 0$, we have

$$\left. \frac{dQ_T^{(k, \theta)}}{dP^{(k)}} \right|_{\mathcal{F}_T^{(k)}} = \frac{1_{A_{k,T}} \left(\prod_{i=1}^k \prod_{u < s_T^{(i)}} L_u \right) e^{-\theta Z_T}}{\mathbb{E} \left[Z_T (Z_T - 1) \cdots (Z_T - k + 1) e^{-\theta Z_T} \right]}$$

filtration of the process including spines

$A_{k,T} = \{ k\text{-spines are alive \& distinct at time } T \}$

$Q_T^{(k, \theta)}$ is a probability measure

$Z_T^{(k)}$ = collection of k -tuples (distinct) of individuals alive at time T

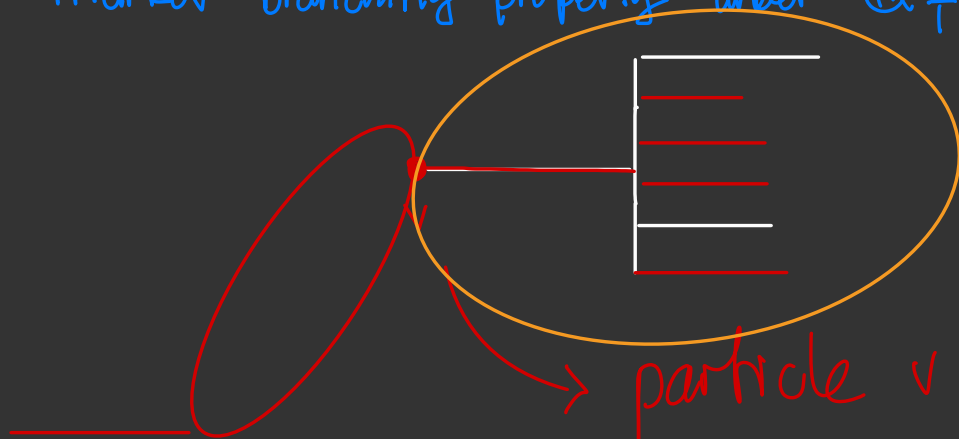
$$|Z_T^{(k)}| = Z_T (Z_T - 1) \cdots (Z_T - k + 1)$$

$$\begin{aligned}\mathbb{E}^{(k)} \left[\mathbb{1}_{A_{k,T}} \left(\prod_{l=1}^k \prod_{u \in \mathcal{Z}_T^{(l)}} L_u \right) e^{-\theta Z_T} \right] &= \sum_{u_1, u_2, \dots, u_k \in \mathcal{Z}_T^{(k)}} \mathbb{P}^{(k)}(\mathcal{Z} = \bar{u} \mid \mathcal{F}_T) \left(\prod_{l=1}^k \prod_{u \in \mathcal{Z}_T^{(l)}} L_u \right) e^{-\theta Z_T} \\ &= e^{-\theta Z_T} \sum_{u_1, u_2, \dots, u_k \in \mathcal{Z}_T^{(k)}} 1 = |\mathcal{Z}_T^{(k)}| e^{-\theta Z_T}\end{aligned}$$

$$\left. \frac{d\mathbb{Q}_T^{(k, \theta)}}{d\mathbb{P}^{(k)}} \right|_{\mathcal{F}_T} = \frac{|\mathcal{Z}_T^{(k)}| e^{-\theta Z_T}}{\mathbb{E}[e^{-\theta Z_T} | \mathcal{Z}_T^{(k)}]}$$

Under $\mathbb{Q}_T^{(k, \theta)}$, the k -spines are a uniform choice without replacement from those individuals alive at time T

Markov branching property under $\mathbb{Q}_T^{(k, \theta)}$



→ particle v with j marks at time t

law of the tree generated $\mathbb{Q}_{T-t}^{(j, \theta)}$

Galton-Watson processes with k -spines under $\mathbb{Q}_T^{(k, \theta)}$ are inhomogeneous multitype GW processes (marks are the types)

$(Z_u, u \in [0, T])$ under $\mathbb{Q}_T^{(k, \theta)}$

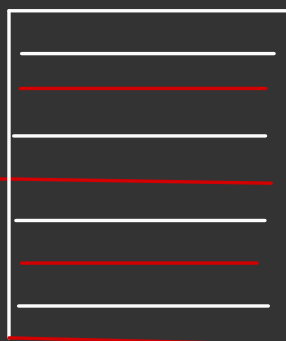
- starts with one particle with k marks
- a particle v with j marks branches into l offspring & the j marks split into $g \in \{1, 2, \dots, j\}$ groups of sizes $k_1, \dots, k_g \geq 1$ ($\sum_{i=1}^g k_i = j$) at rate

$$\frac{j!}{\prod_{m=1}^j h_m \prod_{i=1}^g k_i!} \frac{\prod_{i=1}^g \mathbb{E}[Z_{T-t}^{(Z_{T-t}-1)} (Z_{T-t}-k_i+1) e^{-\theta Z_{T-t}}]}{\mathbb{E}[Z_{T-t}^{(Z_{T-t}-j+1)} e^{-\theta Z_{T-t}}]}$$

$$\times j! (l-1)! (l-g+1)! (\mathbb{E}[e^{-\theta Z_{T-t}}])^{l-g}$$

$h_m = \#\{i \mid k_i = m\}$ $\longleftarrow$ # of marks groups of sizes m

g marks



i) choose g from l $\binom{l}{g}$

ii) assign sizes $k_1, \dots, k_g$ amongs the g $\frac{g!}{\prod_{i=1}^g k_i!}$

iii) partition the g marks between the g chosen off-spring with their given group sizes $\frac{g!}{\prod_{i=1}^g k_i!}$

particles with no marks branches

$$\mathbb{E} \left[e^{\theta Z_{T-t}} \right]^l p_l$$

UNIFORM SAMPLING

f measurable functional on the genealogies

$$\mathbb{E}_{\text{unif}, T}^{(k)}[f(\mathcal{S}_T)] = \mathbb{E}\left[\frac{1}{z_T(z_T-1) \cdots (z_T-k+1)} \sum_{\underline{u} \in \mathcal{Z}^{(k)}} f(\underline{u}) \mid Z_T \geq k\right]$$

$$\mathbb{E}_{\text{unif}, T}^{(k)}[f(\mathcal{S}_T)] = Q_T^{(k, \theta)} \left[\frac{f(\mathcal{S}_T)}{z_T(z_T-1) \cdots (z_T-k+1)} \mathbb{1}_{\{Z_T \geq k\}} \right] \mathbb{E}[z_T(z_T-1) \cdots (z_T-k+1) \mid Z_T \geq k]$$

$0 < t_1 < t_2 < \dots < t_m < T$, $(\beta_0, \beta_1, \dots, \beta_m)$ a splitting process of $[k]$
 with group sizes $g_1, \dots, g_m$ & $l_1, l_2, \dots, l_m$ s.t. $l_i \geq g_i$

$$\mathbb{Q}_T^{(k, \theta)} \left(\tau_1 \in dt_1, \dots, \tau_m \in dt_m, T(S) = (\beta_0, \beta_m), L_{\tau_1} = l_1, \dots, L_{\tau_m} = l_m \right)$$

$$= \frac{F_T'(e^{-\theta})}{\mathbb{E}[Z_T(Z_T-1) \dots (Z_T-k+1) e^{-\theta Z_T}]} \prod_{i=1}^m l_i(l_i-1) \dots (l_i-g_i+1) \mathbb{E}[e^{-\theta Z_{T-t_i}}]^{l_i-g_i} p_{l_i}$$

$$\times \prod_{i=1}^m F_{T-t_i}'(e^{-\theta})^{g_i-1} r dt_i$$

$$F_T(\Delta) = \mathbb{E}[s^{Z_T}]$$

$T \rightarrow \infty$, and use Yaglom's limit

$$\mathbb{E}[\bar{e}^{-\theta \bar{F}(T) Z_T} | Z_T > 0] \xrightarrow{T \rightarrow \infty} 1 - (1 + \theta^{1-\alpha})^{-1/\alpha-1} = \mathbb{E}[\bar{e}^{-\theta W_\alpha}]$$

$$\text{where } \bar{F}(T) = \mathbb{P}(Z_T > 0) \sim c_{\alpha, \gamma} t^{-1/\alpha-1} \frac{1}{l_2(t)}$$

l_2 - slowly varying fct

Thank you !